NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

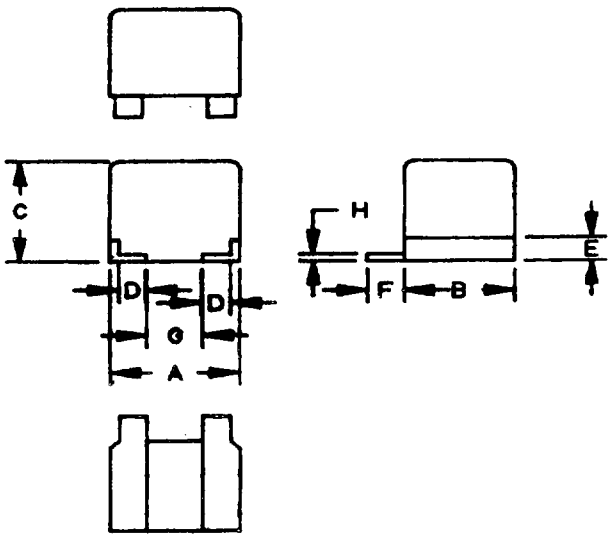
MIL-PRF-83446/10B
20 February 1985
SUPERSEDING
MIL-I-83446/10A
16 December 1980

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, CHIP, FIXED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

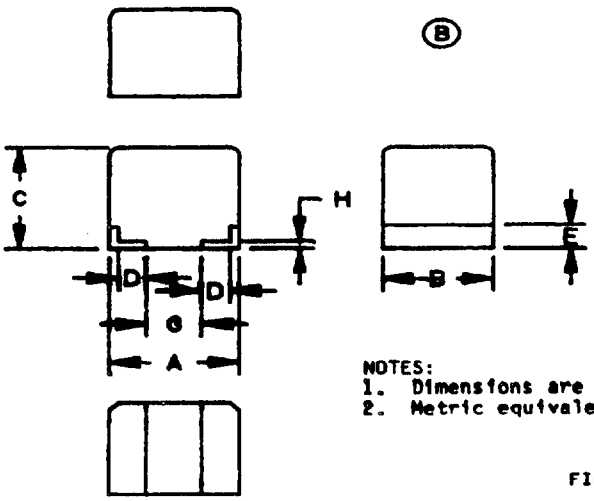
The complete requirements for acquiring the coils described herein shall consist of this specification and the latest issue of MIL-C-83446.




| | INCHES | MM |
|---|---|---|
| A | .155 ±.008 | 3.94 ±0.20 |
| B | .125 ±.008 | 3.18 ±0.20 |
| C | .130 Max | 3.30 Max |
| D | .037 ±.003 | 0.94 ±0.08 |
| E | .020 ±.005 | 0.51 ±0.13 |
| F | .040 +.015 −.010 | 1.02 +0.13 −0.25 |
| G | .073 Min | 1.85 Min |
| H | .005 ±.002 | 0.13 ±0.05 |

For dash numbers 01 thru 61

(WITH TABS)




| | INCHES | MM |
|---|---|---|
| A | .155 ±.008 | 3.94 ±0.20 |
| B | .125 ±.008 | 3.18 ±0.20 |
| C | .130 Max | 3.30 Max |
| D | .037 ±.003 | 0.94 ±0.08 |
| E | .020 ±.005 | 0.51 ±0.13 |
| G | .073 Min | 1.85 Min |
| H | .005 ±.002 | 0.13 ±0.05 |

For dash numbers 62 thru 122

(WITHOUT TABS)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Fixed, chip coil.

Ⓑ denotes changes



FSC 5950

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REQUIREMENTS

Dimensions and configurations: See figure 1.

Weight: 0.5 gram maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C

Altitude: 70,000 feet.

Dielectric withstanding voltage: Method 301 of MIL-STD-202, test voltage 500 volts rms.

Barometric pressure: Method 105, test condition C, MIL-STD-202, test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

TABLE I. Electrical characteristics (initial) and dash numbers.

| Dash number (B) | Inductance ±10% | Q (min) | Q (typ) | Test frequency 1/ | Self-resonant frequency (min) | DC resistance (max) | 2/ Current (max) |
|---|---|---|---|---|---|---|---|
| | uH | | | MHz | MHz | Ohms | mA |
| 01, 62 | 0.010 | 60 | 65 | 150 | 2000 | 0.04 | 1000 |
| 02, 63 | 0.012 | 70 | 82 | " | 1800 | " | " |
| 03, 64 | 0.015 | 75 | 87 | " | 1500 | " | " |
| 04, 65 | 0.018 | 75 | 87 | " | 1500 | " | " |
| 05, 66 | 0.022 | 60 | 65 | 100 | 1300 | 0.05 | " |
| 06, 67 | 0.027 | " | 70 | " | 1300 | 0.05 | " |
| 07, 68 | 0.033 | " | 70 | " | 1000 | 0.05 | " |
| 08, 69 | 0.039 | " | 70 | " | 1000 | 0.06 | 900 |
| 09, 70 | 0.047 | 65 | 75 | " | 800 | 0.06 | 900 |
| 10, 71 | 0.056 | " | " | " | 760 | 0.06 | 900 |
| 11, 72 | 0.068 | " | " | " | 700 | 0.07 | 840 |
| 12, 73 | 0.082 | " | " | " | 650 | " | " |
| 13, 74 | 0.100 | " | 77 | 50 | 570 | " | " |
| 14, 75 | 0.120 | " | 77 | " | 520 | " | " |
| 15, 76 | 0.150 | 75 | 87 | " | 400 | 0.08 | 790 |
| 16, 77 | 0.180 | 75 | 87 | " | 360 | 0.08 | 790 |
| 17, 78 | 0.220 | 70 | 80 | " | 320 | 0.08 | 790 |
| 18, 79 | 0.270 | " | " | " | 270 | 0.10 | 700 |
| 19, 80 | 0.330 | " | " | " | 240 | 0.10 | 700 |
| 20, 81 | 0.390 | " | " | " | 220 | 0.10 | 700 |
| 21, 82 | 0.470 | " | " | 25 | 190 | 0.14 | 590 |
| 22, 83 | 0.560 | " | 82 | " | 170 | 0.19 | 510 |
| 23, 84 | 0.680 | " | 83 | " | 160 | 0.26 | 430 |
| 24, 85 | 0.820 | 75 | 84 | " | 150 | 0.30 | 400 |
| 25, 86 | 1.00 | 75 | 87 | " | 130 | 0.34 | 380 |
| 26, 87 | 1.20 | 65 | 73 | 7.9 | 120 | 0.45 | 330 |
| 27, 88 | 1.50 | " | " | " | 110 | 0.57 | 290 |
| 28, 89 | 1.80 | " | " | " | 100 | 0.72 | 260 |
| 29, 90 | 2.20 | " | " | " | 80 | 0.9 | 230 |

See footnotes at end of table.



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TABLE I. Electrical characteristics (initial) and dash numbers - Continued.

| Dash number Ⓑ | Inductance ±10% | Q (min) | Q (typ) | Test frequency 1/ | Self-resonant frequency (min) | DC resistance (max) | 2/ Current (max) |
|---|---|---|---|---|---|---|---|
| | µH | | | MHz | MHz | Ohms | mA |
| 30, 91 | 2.70 | 65 | 73 | 7.9 | 50 | 1.1 | 210 |
| 31, 92 | 3.30 | 60 | 70 | " | 50 | 1.2 | 200 |
| 32, 93 | 3.90 | 60 | 70 | " | 45 | 1.4 | 180 |
| 33, 94 | 4.70 | 60 | 70 | " | 42 | 1.6 | 170 |
| 34, 95 | 5.60 | 65 | 75 | " | 40 | 1.8 | 160 |
| 35, 96 | 6.80 | " | " | " | 37 | 2.4 | 140 |
| 36, 97 | 8.20 | " | " | " | 34 | 3.0 | 130 |
| 37, 98 | 10.0 | " | " | " | 29 | 3.5 | 120 |
| 38, 99 | 12.0 | 60 | 70 | 2.5 | 27 | 3.6 | 118 |
| 39, 100 | 15.0 | " | 70 | " | 22 | 3.7 | 115 |
| 40, 101 | 18.0 | " | 72 | " | 17 | 3.8 | 114 |
| 41, 102 | 22.0 | " | 72 | " | 16 | 3.9 | 113 |
| 42, 103 | 27.0 | 65 | 75 | " | 15 | 4.0 | 110 |
| 43, 104 | 33.0 | 65 | 75 | " | 14 | 5.0 | 100 |
| 44, 105 | 39.0 | 65 | 75 | " | 13 | 7.0 | 94 |
| 45, 106 | 47.0 | 70 | 78 | " | 12 | 8.0 | 79 |
| 46, 107 | 56.0 | 70 | 78 | " | 11 | 10.0 | 70 |
| 47, 108 | 68.0 | 65 | 75 | " | 10 | 11.0 | 67 |
| 48, 109 | 82.0 | 60 | 72 | " | 9 | 12.0 | 64 |
| 49, 110 | 100.0 | 60 | 70 | " | 8 | 13.0 | 62 |
| 50, 111 | 120.0 | 40 | 48 | 0.79 | 7 | 14.0 | 59 |
| 51, 112 | 150.0 | " | " | " | 6 | 16.0 | 56 |
| 52, 113 | 180.0 | " | " | " | 5 | 18.0 | 52 |
| 53, 114 | 220.0 | " | " | " | 4 | 24.0 | 45 |
| 54, 115 | 270.0 | " | " | " | 3.3 | 25.0 | 44 |
| 55, 116 | 330.0 | 40 | 48 | 0.79 | 3.1 | 29.0 | 41 |
| 56, 117 | 390.0 | 40 | 48 | " | 2.9 | 32.0 | 39 |
| 57, 118 | 470.0 | 35 | 45 | " | 2.4 | 35.0 | 37 |
| 58, 119 | 560.0 | 35 | 45 | " | 2.1 | 45.0 | 33 |
| 59, 120 | 680.0 | 35 | 40 | " | 1.9 | 55.0 | 30 |
| 60, 121 | 820.0 | 30 | 36 | " | 1.8 | 70.0 | 26 |
| 61, 122 | 1000.0 | 30 | 36 | | 1.7 | 80.0 | 25 |

Ⓑ 1/ Test frequency range: 25 through 150 MHz:

For electrical characteristics measurements use TF-250RX-1 test fixture, or equal (see figure 2). Fixture inductance (approximately 0.009 µH) should be subtracted from calculated inductance.)

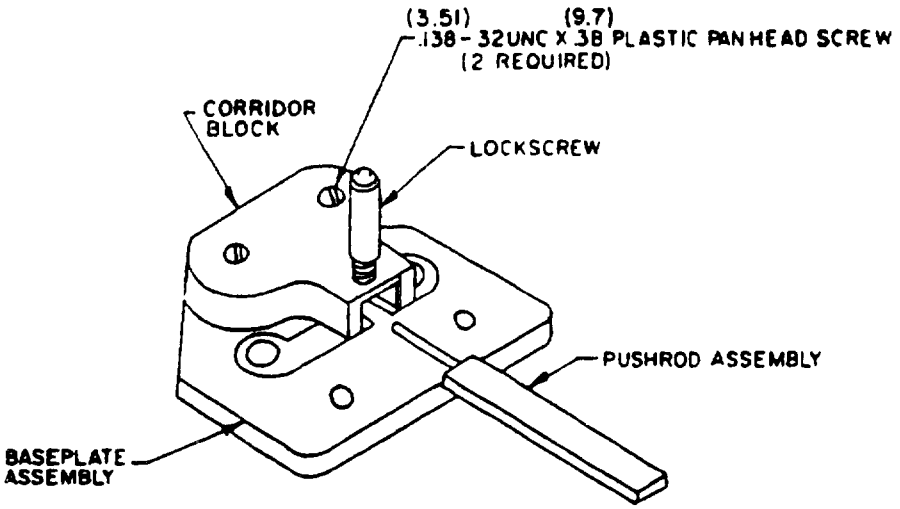
Test frequency range: 0.25 through 25 MHz.:

For electrical characteristics measurements use TF-2600-1 test fixture, or equal (see figure 3). Fixture inductance (approximately 0.028 µH) and residual Q-meter inductance (approximately 0.01 µH) should be subtracted from indicated inductance.

2/ Maximum current allowed not to exceed the specified temperature rise.

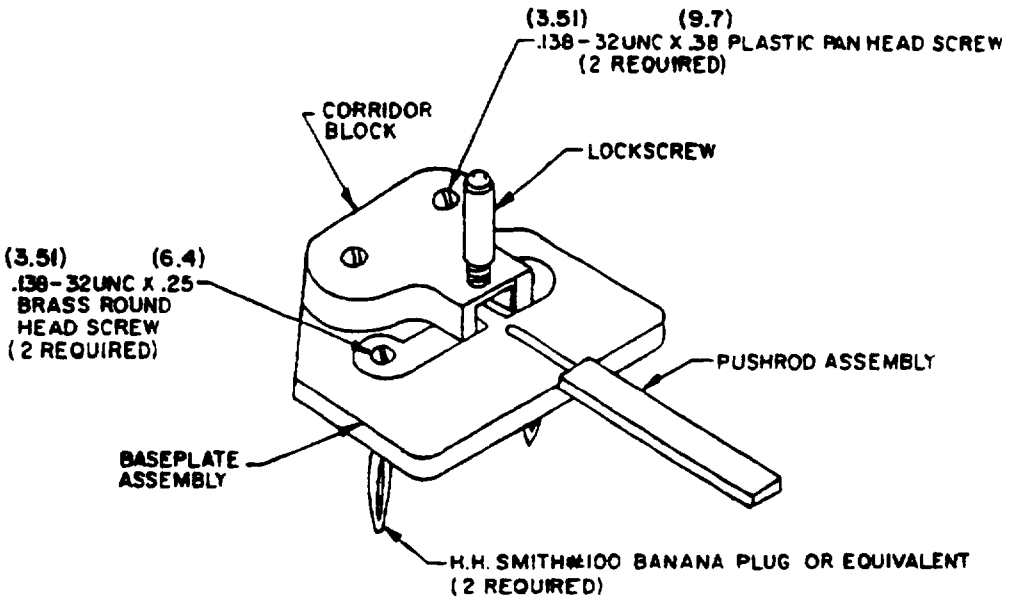


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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 2. Typical chip coil test fixture (TF-250RX-1, or equal).




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 3. Typical chip coil test fixture (TF-2600-1, or equal).

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TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from the initial measurements | | | |
|---|---|---|---|---|
| | Inductance | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | Percent | | Percent | Percent |
| Group II | +5 | +(3% +0.001 ohm) | -8 | -10 |
| Group IV | ∓5 | ∓(2% +0.001 ohm) | -10 | -10 |
| Group V | ∓2 | --- | --- | -10 |
| Quality conformance inspection | | | | |
| Group C | | | | |
| Subgroup II | +5 | +(3% +0.001 ohm) | -9 | -10 |
| Subgroup IV | ∓5 | ∓(3% +0.001 ohm) | -8 | -10 |

Supersession data: A new part numbering system with codes for termination materials was incorporated with revision A, superseding MIL-I-83446/10, dated 1 March 1979.

M83446/10- (dash number from table I) C
supersedes M83446/10- (dash number from table I).

Examples:

M83446/10-21C supersedes M83446/10-21.
M83446/10-03C supersedes M83446/10-3.

Part number: The part number shall be in the following form.

M83446/10- 04 B

04: Sequentially assigned dash numbers (see table I)

B: Termination finish (see MIL-C-83446)

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS
Air Force - 19
DLA - ES

User activities:
Navy - OS
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5950-0633-5)

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